Quarry: Technical Architecture Specification

Local Semantic Search for AI Agents

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Contents

1 Overview **3**
- 1.1 Design Principles . . . 3

2 System Architecture **3**
- 2.1 Deployment Topology . . . 3
- 2.2 Surface Summary . . . 3

3 Module Responsibilities **4**
- 3.1 Data Layer . . . 4
- 3.2 Ingestion Pipeline . . . 4
- 3.3 Format Processors . . . 4
- 3.4 Backends . . . 5
- 3.5 Sync and Registry . . . 5
- 3.6 Surfaces . . . 5
- 3.7 Support . . . 5

4 Daemon Model **5**
- 4.1 Why a Daemon . . . 5
- 4.2 mcp-proxy Architecture . . . 6
- 4.3 CLI Independence . . . 6

5 MCP Tool Surface **6**

6 HTTP Server **6**
- 6.1 ASGI Application . . . 6
- 6.2 Endpoints . . . 7
- 6.3 Port File . . . 7

7 Configuration **7**
- 7.1 Settings . . . 7
- 7.2 Named Databases . . . 7
- 7.3 Chunking Tuning . . . 8

8 Search and Retrieval **8**
- 8.1 Embedding Model . . . 8
- 8.2 OCR . . . 8
- 8.3 Search Quality Tips . . . 8
- 8.4 Hybrid Search . . . 9
- 8.5 Agent Memory Schema . . . 9
- 8.6 Ethos Extension Integration . . . 9

9 Logging and Exceptions **10**
- 9.1 Logging Standard . . . 10
- 9.2 Exception Handling . . . 10

10 Security **10**
10.1 Bearer Token Authentication . . . 10
10.2 WebSocket Security (CSWSH) . . . 11
10.3 CORS . . . 11
10.4 TLS and TOFU Certificate Pinning . . . 11
10.5 Log Safety (CWE-532, CWE-117) . . . 11

11 Deployment **12**
11.1 Single Machine . . . 12
11.2 Network Mode (Remote GPU) . . . 12
11.3 Client (Remote Access) . . . 12
11.4 Daemon (`quarry serve`) . . . 12
11.5 Container (Fly.io) . . . 12

12 Performance **12**
12.1 Critical Path . . . 12
12.2 Fire-and-Forget MCP Pattern . . . 13
12.3 Single Table Design . . . 13

13 Plugin and Hook System **13**
13.1 Session Start . . . 13
13.2 Web Fetch Capture . . . 13
13.3 Pre-Compact Capture . . . 13

14 CLI Commands **13**

15 Test Architecture **14**

16 Formal Model **15**

A Operation Concurrency Model **16**
A.1 Concurrency Matrix . . . 16
A.2 Design Rationale . . . 16
A.3 Task Lifecycle . . . 17

1 Overview

Quarry is a local semantic search engine. It ingests documents in 20+ formats, embeds them with a local ONNX model, stores vectors in LanceDB, and serves semantic search to AI agents (Claude Code, Claude Desktop) and a macOS menu bar app.

The system follows the Punt Labs projection strategy: a single core library projects outward through four thin adapter layers. Each surface imports core functions and calls them with resolved arguments. No intermediary service layer.

1.1 Design Principles

P1. **Library-first.** The core library does all work. CLI, MCP, HTTP, and plugin surfaces are thin wrappers that resolve settings and delegate.

P2. **Local by default.** Everything runs on the user's machine. No API keys, no cloud accounts. The embedding model downloads once (∼120 MB, int8 quantized).

P3. **One daemon, many clients.** A single `quarry serve` process loads the embedding model once and serves all sessions over WebSocket. Clients connect via `mcp-proxy`.

P4. **Fire-and-forget I/O.** Side-effect MCP tools return immediately and process in a bounded thread pool. Search is synchronous.

P5. **Named databases.** Each database is fully isolated: its own LanceDB directory, sync registry, and vector index. Work/personal separation without configuration files.

2 System Architecture

2.1 Deployment Topology

Single machine (localhost TLS):

```
                  stdio wss:// (TLS)
Claude Code <------------------> mcp-proxy <--------------------> quarry serve
            MCP JSON-RPC (~5 MB Go) pinned CA cert (one daemon)
```

Remote server (GPU):

```
Mac (client) Linux (server)
                  stdio wss:// (TLS)
Claude Code <------------------> mcp-proxy <----------------------> quarry serve
            MCP JSON-RPC (~5 MB Go) pinned CA cert (GPU daemon)

CLI -----> HTTPS GET/POST ---------> quarry serve
          Bearer + pinned CA
```

Without the proxy, every Claude Code tab spawns a separate Python process, each loading the embedding model into ∼200 MB of RAM. With it, startup is instant and state is shared across all sessions. Installed services use TLS with TOFU certificate pinning (DES-020) — even on localhost.

When a remote server is configured (DES-021), the CLI routes all data commands over HTTPS to the server, with only authentication and administration commands remaining local. As of v1.12.4, `find`, `show`, `status`, `ingest`, `remember`, `delete`, `sync`, and all `list` subcommands route remotely. Registration create/delete and `use` are the remaining gaps.

2.2 Surface Summary

Surface	Transport	Purpose
CLI	Direct library calls	Interactive use from terminal
MCP server	stdio or WebSocket	Claude Code/Desktop tools
HTTP server	REST JSON + WS `/mcp`	Menu bar app + mcp-proxy
Plugin	Hook shell commands	Auto-register, auto-ingest, compaction safety

3 Module Responsibilities

3.1 Data Layer

Module	File	Responsibility
Models	`src/quarry/models.py`	Immutable dataclasses: `PageType` enum, `PageContent`, `Chunk`, `PageAnalysis`. All `@dataclass(frozen=True)`.
Types	`src/quarry/types.py`	Protocol definitions for external dependencies: `LanceDB`, `LanceTable`, `LanceQuery`, `ListTablesResult`, `OcrBackend`, `EmbeddingBackend`. Structural typing instead of `Any`.
Config	`src/quarry/config.py`	`Settings` (pydantic `BaseSettings`), `resolve_db_paths()` for named databases, ONNX model constants.
Database	`src/quarry/database.py`	LanceDB operations: `insert_chunks`, `search`, `list_documents`, `delete_document`, `count_chunks`, etc. Single table `chunks`.

3.2 Ingestion Pipeline

Module	File	Responsibility
Pipeline	`src/quarry/pipeline.py`	Orchestrator: dispatches by format, chunks, embeds, stores. Entry points: `ingest_document`, `ingest_content`, `ingest_auto` (URL with sitemap discovery).
Chunker	`src/quarry/chunker.py`	Text splitting with overlap. Respects page boundaries and heading structure.
Collections	`src/quarry/collections.py`	`derive_collection()` — determines collection name from path or URL.

3.3 Format Processors

Each processor converts a source format into `list[PageContent]`:

Module	File	Formats
PDF	`src/quarry/pdf_analyzer.py`	PDF (text via PyMuPDF, OCR fallback for image pages)
Image	`src/quarry/image_analyzer.py`	PNG, JPG, TIFF, BMP, WebP (OCR)
Text	`src/quarry/text_processor.py`	TXT, MD, LaTeX, DOCX
Code	`src/quarry/code_processor.py`	30+ languages (AST-aware splitting)
HTML	`src/quarry/html_processor.py`	HTML (boilerplate stripping, Markdown conversion)
Spreadsheet	`src/quarry/spreadsheet_processor.py`	XLSX, CSV (tabular serialization)
Presentation	`src/quarry/presentation_processor.py`	PPTX (slide-per-chunk with notes)
Dispatcher	`src/quarry/text_extractor.py`	Extracts text from text-based PDF pages (PyMuPDF)

3.4 Backends

Module	File	Responsibility
Factory	`src/quarry/backends.py`	Thread-safe factory with double-checked locking. `get_ocr_backend()` and `get_embedding_backend()` return cached singletons.
OCR	`src/quarry/ocr_local.py`	Local OCR via RapidOCR (offline, no setup).
Embeddings	`src/quarry/embeddings.py`	Local ONNX embedding (snowflake-arctic-embed-m-v1.5, 768-dim, 512 tokens).

3.5 Sync and Registry

Module	File	Responsibility
Sync	`src/quarry/sync.py`	Directory sync: discover files, compute delta (new/changed/deleted), ingest/delete accordingly. `ThreadPoolExecutor` for concurrent ingestion.
Registry	`src/quarry/sync_registry.py`	SQLite registry of watched directories. Maps directories to collections with timestamps.

3.6 Surfaces

Module	File	Responsibility
CLI	`src/quarry/__main__.py`	Typer CLI with rich formatting. Synchronous. All commands: resolve settings → call core → format output.
MCP Server	`src/quarry/mcp_server.py`	FastMCP server. Read-only tools are synchronous. Side-effect tools use fire-and-forget.
HTTP Server	`src/quarry/http_server.py`	Starlette ASGI app on uvicorn. REST endpoints + WebSocket `/mcp`.
Hooks	`src/quarry/hooks.py`	Claude Code plugin hooks: session-start (auto-register), post-web-fetch (auto-ingest), pre-compact (capture transcript). All fail-open.

3.7 Support

Module	File	Responsibility
Formatting	`src/quarry/formatting.py`	Pre-formatted plain text output using unicode box-drawing. Used by both CLI and MCP for consistent output.
Results	`src/quarry/results.py`	`SyncResult` dataclass for sync operation reporting.
Sitemap	`src/quarry/sitemap.py`	Sitemap discovery and URL extraction for web ingestion.
Doctor	`src/quarry/doctor.py`	Health checks and install flow: model availability, database access, MCP configuration, ethos extension setup.

4 Daemon Model

4.1 Why a Daemon

The embedding model (snowflake-arctic-embed-m-v1.5, int8 quantized ONNX) occupies ∼200 MB of RAM and takes ∼250 ms to load. MCP's stdio transport model spawns a new process per session.

Without a daemon, five Claude Code tabs means five copies of the model in memory and five cold starts.

`quarry serve` loads the model once and keeps it resident. The `mcp-proxy` binary (∼5 MB Go binary, <10 ms startup) bridges each session's MCP stdio to the daemon over WebSocket.

4.2 mcp-proxy Architecture

`quarry install` downloads `mcp-proxy` (SHA256-verified, platform-specific binary) and configures MCP clients to use it. The proxy:

1. Reads MCP JSON-RPC from stdin.
2. Forwards to `ws://localhost:8420/mcp` over WebSocket.
3. Relays responses back to stdout.

Each WebSocket connection gets its own asyncio Task with a `ContextVar` for database selection (`_db_name`), so `use("work")` in one session does not affect others.

4.3 CLI Independence

The CLI does not use the daemon. `quarry find "margins"` loads the embedding model in-process, queries LanceDB, prints results, and exits. LanceDB handles concurrent reads safely, so CLI and daemon can access the same database simultaneously.

5 MCP Tool Surface

Tool	Key Parameters	Purpose	Exec
`find`	`query`, `limit`, `collection`	Semantic search	Sync
`ingest`	`source`, `overwrite`, `collection`	Ingest file or URL	Bg
`remember`	`content`, `document_name`	Ingest inline text	Bg
`list`	`kind`	List indexed data	Sync
`show`	`document_name`, `page_number`	Document metadata or page	Sync
`delete`	`name`, `kind`	Delete document or collection	Bg
`register_directory`	`directory`, `collection`	Register for sync	Bg
`deregister_directory`	`collection`, `keep_data`	Remove registration	Bg
`sync_all`	—	Sync all registered dirs	Bg
`status`	—	Database stats	Sync
`use`	`name`	Switch active database	Sync

Bg = background (`ThreadPoolExecutor`).

Background tools use a bounded `ThreadPoolExecutor(max_workers=4)`. They return an optimistic response immediately. Settings and database connections are snapshotted at the tool boundary and passed into background functions — this avoids a race condition where the mutable `_db_name` ContextVar could change between the tool call and the background thread's execution. Exceptions in background threads are logged via `logger.exception()`, not raised.

6 HTTP Server

6.1 ASGI Application

Starlette app built by `build_app()` factory, served by uvicorn. REST handlers are sync functions — Starlette auto-detects this and runs them in its threadpool. The `/mcp` WebSocket endpoint is async.

Default port: 8420 (`DEFAULT_PORT` in `src/quarry/config.py`), overridable with `--port`.

6.2 Endpoints

Path	Method	Purpose	Auth	Model
/health	GET	Server uptime and status	No	sync
/ca.crt	GET	CA certificate PEM (TOFU)	No	sync
/search	GET	Hybrid search	Yes	sync
/show	GET	Document metadata or page text	Yes	sync
/documents	GET, DELETE	List or delete documents	Yes	sync
/collections	GET, DELETE	List or delete collections	Yes	sync
/registrations	GET, POST, DELETE	Manage directory registrations	Yes	sync
/databases	GET	List named databases	Yes	sync
/status	GET	Database statistics	Yes	sync
/use	POST	Switch active database	Yes	sync
/remember	POST	Ingest inline content	Yes	sync
/ingest	POST	Ingest URL (returns 202)	Yes	async
/ingest/{id}	GET	Poll ingest task status	Yes	—
/sync	POST	Sync registrations (returns 202)	Yes	async
/sync/{id}	GET	Poll sync task status	Yes	—
/mcp	WS	MCP JSON-RPC over WebSocket	Yes	—

See Appendix A for the full concurrency matrix across HTTP, CLI, and MCP surfaces.

6.3 Port File

After binding, the server writes the actual port to `~/.punt-labs/quarry/data/<db>/serve.port`. This supports `--port 0` (ephemeral assignment). Removed on shutdown via lifespan context.

7 Configuration

7.1 Settings

All settings are environment variables read via pydantic-settings.

Variable	Default	Description
Paths		
QUARRY_ROOT	~/.punt-labs/quarry/data	Base directory for all databases
LANCEDB_PATH	<root>/default/lancedb	Vector database location
REGISTRY_PATH	<root>/default/registry.db	Directory sync SQLite database
LOG_PATH	~/.punt-labs/quarry/logs/quarry.log	Log file (rotating, 5 MB, 3 backups; independent of QUARRY_ROOT)
Chunking		
CHUNK_MAX_CHARS	1800	Target max characters per chunk (∼450 tokens)
CHUNK_OVERLAP_CHARS	200	Overlap between consecutive chunks
Embedding		
EMBEDDING_MODEL	Snowflake/snowflake-arctic-embed-m-v1.5	Model identifier (display only — the ONNX model is fixed)
EMBEDDING_DIMENSION	768	Vector dimension (display only — fixed at 768)

7.2 Named Databases

`resolve_db_paths(settings, name)` maps a database name to a path under `QUARRY_ROOT`. Each database is fully isolated — its own LanceDB directory, sync registry, and vector index. Names containing path

separators (/ or \) or the literal traversal segments `.` or `..` are rejected; names like `foo.bar` are allowed. If `LANCEDB_PATH` was explicitly overridden, the override is preserved.

The persistent default database is stored in `~/.punt-labs/quarry/config.toml` as `[default] database = "name"`.

7.3 Chunking Tuning

Increase `CHUNK_MAX_CHARS` (3000–4000) for documents with long self-contained sections: legal contracts, academic papers, large source code functions.

Decrease `CHUNK_MAX_CHARS` (800–1200) for dense documents where each paragraph covers a distinct topic.

Increase `CHUNK_OVERLAP_CHARS` (400) when relevant information spans paragraph boundaries.

The embedding model has a 512-token context window. Chunks exceeding this are truncated during embedding, reducing quality for the tail.

8 Search and Retrieval

8.1 Embedding Model

snowflake-arctic-embed-m-v1.5 via ONNX Runtime:

- 768-dimensional vectors, 512-token context window
- Asymmetric retrieval: queries are prefixed with a search instruction, documents are not
- int8 quantized ONNX for CPU (∼120 MB), FP16 for CUDA (∼220 MB)
- Provider auto-detection: CUDA+FP16 when available, CPU+int8 fallback. `QUARRY_PROVIDER` env var overrides (`cpu`, `cuda`, or unset for auto). See DES-016 and `provider.py`.
- Auto-downloads on first use via `huggingface_hub`

The model is fixed — there is no configuration to swap it. Changing the model invalidates all existing embeddings, requiring full re-ingestion.

8.2 OCR

RapidOCR handles all image-based text extraction. For semantic search, OCR accuracy matters less than expected — the embedding model is robust to minor OCR errors. A misspelled word rarely changes semantic meaning enough to affect search ranking.

8.3 Search Quality Tips

1. **Use natural language queries.** The model is trained on question-passage pairs. "What were Q3 revenue figures?" works better than "Q3 revenue."
2. **Use collection filtering.** `quarry find "auth logic" -c backend`.
3. **Re-ingest after tuning.** Chunk parameters only affect new ingestions. Run `quarry sync` after changes.
4. **Check chunk boundaries.** `quarry find -n 1 "query"` and examine the result.
5. **Use `--overwrite` when re-ingesting.** Without it, duplicate chunks accumulate.

8.4 Hybrid Search

When an `agent_handle` filter is provided, quarry uses dual-channel retrieval instead of vector-only search.

Channels:

1. **Vector similarity** — existing ANN search via LanceDB
2. **BM25 full-text search** — Tantivy index on the `text` column, created during schema migration

Results are fused using Reciprocal Rank Fusion (RRF):

$$\text{score}(m) = \sum_{\text{channel}} \frac{1}{60 + \text{rank}_{\text{channel}}(m)}$$

Temporal decay applies to agent-scoped memories (non-empty `agent_handle`):

$$w = e^{-\lambda \cdot h}$$

where λ is the decay rate and h is hours since ingestion. Unscoped documents (`agent_handle` = “”) are exempt from decay.

When no `agent_handle` is provided, search uses the existing vector-only path (backwards compatible).

8.5 Agent Memory Schema

Three columns extend the chunks table for agent-scoped memory:

Column	Type	Default	Purpose
`agent_handle`	utf8	`""`	Agent identity scope
`memory_type`	utf8	`""`	fact, observation, opinion, procedure
`summary`	utf8	`""`	One-line summary

Migration is idempotent — `_migrate_schema()` adds missing columns with empty-string defaults on every table open. Existing databases gain the columns transparently.

Identity tagging: the PreCompact hook reads the agent handle from `.punt-labs/ethos/config.yaml` and tags captured transcripts automatically.

8.6 Ethos Extension Integration

Quarry stores its configuration in ethos’s generic extension mechanism (DES-008, DES-019):

```
~/.punt-labs/ethos/identities/<handle>.ext/quarry.yaml
```

The extension file contains two categories of data:

1. **Sidecar config** — `memory_collection` and `expertise_collections` are read directly by quarry via the filesystem (no ethos API call).
2. **Session context** — `session_context` is a YAML literal block scalar emitted verbatim by ethos at session start and before context compaction. It contains the agent’s memory instructions (collection name, slash commands, memory types).

`quarry install` (step 7/7) scans all identity extension directories and appends `session_context` to any `quarry.yaml` that has `memory_collection` but no `session_context`. Uses raw file append to preserve existing YAML comments and formatting. Per-identity exception handling ensures one malformed file does not abort the rest.

Ownership boundary: ethos emits quarry’s instructions without understanding them. Quarry writes its own instructions without importing ethos. The dependency is one-way — quarry depends on ethos for identity; ethos has zero knowledge of quarry’s internals.

9 Logging and Exceptions

9.1 Logging Standard

Every module that performs I/O declares `logger = logging.getLogger(__name__)`. Pure data modules (`models.py`, `types.py`, `chunker.py`) do not.

Logging is configured once at the application entry point (CLI or MCP server). Library modules never call `logging.basicConfig`.

Level	Use
`DEBUG`	Internal state: variable values, branch decisions, intermediate results.
`INFO`	Operational milestones: starting a job, completing a step, resource counts.
`WARNING`	Unexpected but recoverable: empty input, deprecated parameter usage.
`ERROR`	Failures that prevent completion but do not crash. Always with exception context.

`CRITICAL` is not used. If the process must exit, it raises an exception.

Log messages use `%s`-style formatting (lazy evaluation), not f-strings. Each message includes enough context to trace an operation without reading code.

9.2 Exception Handling

The codebase uses built-in exception types. No custom exceptions unless a caller needs to distinguish failure modes programmatically.

Exception	Raised when
`ValueError`	Invalid input: unsupported format, bad parameter.
`FileNotFoundError`	A file path argument does not exist.
`RuntimeError`	An operation reports failure (OCR engine error, ingestion failure).
`TimeoutError`	An external service exceeds its time budget.

Rules:

1. Never swallow exceptions. Every `except` block must re-raise, or log with `logger.exception()` and raise a different exception, or (at system boundaries only) return a documented sentinel.
2. Never use bare `except:`.
3. Never catch `Exception` broadly in library code. Broad catches are permitted only at the outermost boundary (MCP tool handler, CLI command).
4. Exceptions carry context: include the values that caused the failure.
5. Use `try/finally` for cleanup, not `try/except`.
6. Do not use exceptions for flow control.

10 Security

10.1 Bearer Token Authentication

`quarry serve --api-key` (or `QUARRY_API_KEY` env var) gates all HTTP endpoints except `/health` behind `Authorization: Bearer <key>`.

- `/health` is exempt — load balancers need unauthenticated health checks.

- `OPTIONS` is exempt — CORS preflight requests carry no auth.
- `hmac.compare_digest` for token comparison — prevents timing attacks.
- Case-insensitive scheme per RFC 7235.
- Empty key = no auth — prevents accidental trivial bypass.
- No key = auth disabled — local use requires no key.

The server refuses to bind to a non-loopback address without `--api-key`.

10.2 WebSocket Security (CSWSH)

Origin-based cross-site WebSocket hijacking protection:

- If an `Origin` header is present, it must match the CORS allow-list.
- Browsers always send `Origin`; non-browser clients like `mcp-proxy` do not.
- Bearer auth is checked before WebSocket accept (close code 1008 on failure).

10.3 CORS

`CORSMiddleware` configured with GET and OPTIONS methods, Authorization and Content-Type headers. Default allowed origin: `http://localhost`. Configurable via `--cors-origin` (repeatable).

The server reflects the request's `Origin` header only when it matches the allow list, and adds `Vary: Origin` to signal cacheability varies by origin. When no `Origin` is present, no CORS headers are emitted.

10.4 TLS and TOFU Certificate Pinning

The installed service always runs with TLS enabled. `quarry serve` without `--tls` is supported for local development but must not be used for production. Certificate management uses TOFU (Trust On First Use) with self-signed CA pinning (DES-020).

Certificate generation: `quarry install` generates an EC P-256 CA and server certificate. Files are written atomically to `~/.punt-labs/quarry/tls/` with restricted permissions (0600 for keys, 0644 for CA cert). The server cert SAN includes the configured hostname (via `QUARRY_TLS_HOSTNAME`), localhost, and loopback addresses.

TOFU login:

1. Client fetches `/ca.crt` over HTTPS (verification disabled — bootstrap).
2. Displays SHA256 fingerprint for out-of-band confirmation.
3. Pins CA cert at `~/.punt-labs/mcp-proxy/quarry-ca.crt`.
4. All subsequent connections verify against the pinned CA only — system roots excluded.

mcp-proxy: The `quarry.toml` profile includes `ca_cert` pointing to the pinned CA. mcp-proxy builds a custom `tls.Config` with the pinned CA as the sole root, enforcing TLS 1.3 minimum.

10.5 Log Safety (CWE-532, CWE-117)

Uvicorn's access log is disabled (`access_log=False`), eliminating query string leakage at the source. The search handler logs only result count, never the raw query. WebSocket session keys are sanitized before logging: control characters stripped via `_CONTROL_CHAR_RE` and truncated to 64 characters to prevent log injection (CWE-117) and log flooding.

11 Deployment

11.1 Single Machine

```
curl -fsSL https://raw.githubusercontent.com/.../install.sh | sh
```

Installs quarry, embedding model, mcp-proxy, Claude Code plugin, and registers a daemon with TLS on localhost.

11.2 Network Mode (Remote GPU)

```
export QUARRY_API_KEY=$(openssl rand -hex 32)
curl -fsSL https://raw.githubusercontent.com/.../install.sh | sh -s -- --network
```

Same as default, but binds daemon to `0.0.0.0:8420` instead of localhost. Requires `QUARRY_API_KEY`. Installs quarry, embedding model, generates TLS certificates, registers systemd service. Detects NVIDIA GPUs and swaps `onnxruntime` for `onnxruntime-gpu`. Prints CA fingerprint for client TOFU. Claude Code plugin installed if `claude` CLI is on PATH.

11.3 Client (Remote Access)

```
curl -fsSL https://raw.githubusercontent.com/.../install.sh | sh
quarry login <server> --api-key <token>
```

No special flag needed. Default install runs a local daemon on localhost. `quarry login` redirects queries to the remote server via TOFU certificate pinning and writes the mcp-proxy config.

11.4 Daemon (`quarry serve`)

```
QUARRY_API_KEY=<token> quarry serve --port 8420 --host 0.0.0.0 --tls
```

Non-loopback hosts require an API key (the server refuses to start without one).

Managed by `launchd` (macOS) or `systemd` (Linux) via `quarry install`. API key delivered via systemd `EnvironmentFile` (Linux) or launchd `EnvironmentVariables` (macOS) — never in process arguments.

11.5 Container (Fly.io)

Multi-stage Dockerfile, three stages:

1. **deps** (`python:3.13-slim`): installs `uv`, runs `uv sync --frozen --no-dev`. Two-pass for Docker layer caching (deps first, then source).
2. **model** (extends `deps`): downloads the embedding model at build time (∼120 MB, int8 quantized). Cached as a separate Docker layer.
3. **runtime** (`python:3.13-slim`): copies `/app` from deps, copies model cache from model stage. `QUARRY_ROOT=/data`, port 8080.

LanceDB data lives on a Fly persistent volume at `/data`. `QUARRY_API_KEY` is set via `fly secrets`. Cold start is ∼5 s with the baked-in model.

12 Performance

12.1 Critical Path

Operation	Latency	Notes
Model load	∼250 ms	One-time per process
Embedding (1 chunk)	∼50 ms	Apple Silicon, CPU
LanceDB search	<10 ms	Vector similarity + filter
End-to-end search	<100 ms	After model is loaded

12.2 Fire-and-Forget MCP Pattern

Side-effect MCP tools return immediately and process in a bounded thread pool (4 workers). This is necessary because:

1. MCP tool calls block the LLM's response stream. A 30-second ingest blocks Claude from responding.
2. The bounded pool prevents resource exhaustion under concurrent requests.
3. Settings and database connections are snapshotted at the tool boundary, avoiding race conditions with the mutable `_db_name` ContextVar.

12.3 Single Table Design

All chunks live in one LanceDB table (`chunks`). Document and collection boundaries are columns, not separate tables. This simplifies cross-document search and avoids table proliferation. Filtering by document/collection/page_type/source_format uses LanceDB's built-in filter predicates on the vector search.

13 Plugin and Hook System

The Claude Code plugin (`.claude-plugin/`) registers hooks that fire on session events and tool calls. All hooks are fail-open: exceptions are logged and the process exits 0.

Hook Event	Matcher	Effect	Blocks?
SessionStart	*	Auto-register directory + background sync	No
PostToolUse	quarry tools	Suppress tool output formatting	No
PostToolUse	WebFetch	Auto-ingest fetched URL	No
PreCompact	*	Capture transcript as session notes	No

13.1 Session Start

`handle_session_start()` auto-registers the working directory and launches a background sync via a detached subprocess. Uses atomic PID-file locking (`O_CREAT|O_EXCL`) at `~/.punt-labs/quarry/sync.pid` to prevent concurrent syncs. Returns `additionalContext` immediately without waiting.

13.2 Web Fetch Capture

`handle_post_web_fetch()` ingests URLs fetched during a session into the `web-captures` collection. Deduplicates by document name. Prefers already-fetched content from the tool response (avoids refetch, reduces SSRF surface).

13.3 Pre-Compact Capture

`handle_pre_compact()` reads the transcript (JSONL) and ingests it into the `session-notes` collection. Extracts user/assistant message text, skipping tool-use blocks. Capped at 500,000 characters.

14 CLI Commands

Command	Purpose
`ingest <source>`	Ingest file or URL. Flags: `--agent-handle`, `--memory-type`, `--summary`
`remember`	Ingest stdin content
`find <query>`	Semantic search

Command	Purpose
`show <document>`	Document metadata or page text
`status`	Database statistics
`use <name>`	Set persistent default database
`delete <name>`	Delete document or collection
`register <directory>`	Register for incremental sync
`deregister <collection>`	Remove directory registration
`sync`	Sync all registered directories
`list documents`	List indexed documents
`list collections`	List collections with counts
`list databases`	List named databases
`list registrations`	List registered directories
`serve`	Start HTTP API server
`mcp`	Start MCP server (stdio)
`login <host>`	TOFU login to remote server (cert pinning)
`logout`	Disconnect from remote server
`remote list`	Show remote config; `--ping` validates connection
`install`	Set up data directory, download model, generate TLS certs, register daemon, detect GPU, configure ethos ext
`doctor`	Check environment health
`uninstall`	Remove system daemon
`version`	Print version

Global flags: `--json`, `--verbose/-v`, `--quiet/-q`, `--db <name>`.

Remote routing (DES-021): When a remote server is configured (via `quarry login`), all data commands should route to the remote server over HTTPS with the pinned CA cert. Only authentication commands (`login`, `logout`, `remote`) and administration commands (`install`, `uninstall`, `serve`, `mcp`, `version`) are local-only. As of v1.12.4, all data commands (`find`, `show`, `status`, `ingest`, `remember`, `delete`, `sync`, and all `list` subcommands) route remotely. Registration create/delete and `use` remain local-only.

15 Test Architecture

821 tests across 33 files. Three tiers plus integration:

Tier	Strategy	Files
Tier 1: Core logic	Real data, no mocks	`test_database`, `test_chunker`, `test_embeddings`, `test_code_processor`, `test_html_processor`, `test_text_processor`, `test_collections`, `test_config`, `test_formatting`, `test_image_analyzer`, `test_latex_utils`, `test_models`, `test_pdf_analyzer`, `test_presentation_processor`, `test_spreadsheet_processor`, `test_text_extractor`, `test_registry` (17 files)
Tier 2: Orchestration	Mocked I/O	`test_pipeline`, `test_pipeline_images`, `test_ocr_local`, `test_url_ingestion`, `test_sitemap`, `test_backends`, `test_sync`, `test_hooks`, `test_doctor` (9 files)
Tier 3: Surface wiring	Fully mocked, <2 s	`test_cli`, `test_mcp_server`, `test_mcp_websocket`, `test_http_server`, `test_proxy`, `test_service` (6 files)
Integration	Real ONNX + real LanceDB	`test_integration` (`@pytest.mark.slow`)

Quality gates: `ruff` (lint + format), `mypy` (strict), `pyright` (strict), `pytest` (all markers except slow/integration). Zero violations required.

16 Formal Model

The plugin state machine — how quarry hooks into the Claude Code session lifecycle — is formally specified in `docs/claude-code-quarry.tex` using Z notation (ISO 13568).

The specification covers:

- Session start with/without auto-sync
- Web fetch auto-capture into `web-captures` collection
- Pre-compaction transcript capture into `session-notes`
- Database switching via `use` tool
- Knowledge hint injection at tool boundaries
- Compaction observability: a capture flag records whether capture preceded compaction; the hook system enforces capture-before-compaction on a best-effort basis
- Clean session boundaries: all quarry state cleared on `EndSession`

The specification can be type-checked with `fuzz -t claude-code-quarry.tex`.

A Operation Concurrency Model

Every quarry operation is exposed on up to three surfaces: the HTTP API (daemon), the CLI, and the MCP server (stdio or WebSocket). The concurrency model differs by surface because each has different constraints: HTTP connections time out, MCP tool calls block the LLM response stream, and CLI commands run in a user's terminal.

Table 4 classifies each operation. The **Client model** column describes what the caller experiences:

sync The caller blocks until the operation completes and receives the result inline.

fire-and-forget The caller receives an acknowledgement immediately (HTTP 202 or a short string). Work runs in the background; the caller polls or ignores the outcome.

— The operation is not available on that surface.

A.1 Concurrency Matrix

Table 4: Operation concurrency by surface. *Long-running* = involves network I/O, embedding, or bulk LanceDB writes.

Operation	HTTP API	CLI (local)	CLI (remote)	MCP
Read operations				
find (search)	sync	sync	sync	sync
show	sync	sync	sync	sync
status	sync	sync	sync	sync
list documents	sync	sync	sync	sync
list collections	sync	sync	sync	sync
list registrations	sync	sync	sync	sync
list databases	sync	sync	sync	sync
health	sync	—	—	—
Short write operations				
register	sync	sync	sync	fire-and-forget
deregister	sync	sync	sync	fire-and-forget
delete (document)	sync	sync	sync	fire-and-forget
delete (collection)	sync	sync	sync	fire-and-forget
use (switch db)	sync	sync	sync	sync
Long-running operations				
ingest (URL)	fire-and-forget	sync	fire-and-forget	fire-and-forget
ingest (file)	—	sync	—	fire-and-forget
remember	sync	sync	sync	fire-and-forget
sync	fire-and-forget	sync	fire-and-forget	fire-and-forget
optimize	—	sync	—	—

A.2 Design Rationale

Read operations are synchronous on all surfaces. Hybrid search (embedding + vector similarity + BM25) completes in <100 ms after model load; the overhead of task tracking would exceed the operation itself.

Short writes (register, deregister, delete) are synchronous on HTTP and CLI because they complete in milliseconds. MCP uses fire-and-forget to avoid blocking the LLM response stream, even for fast operations — the bounded thread pool (4 workers) is cheap insurance.

Long-running operations must be fire-and-forget on HTTP because the CLI client has a finite connection timeout. Sitemap crawls, multi-file syncs, and large-document embedding routinely exceed 15 seconds.

- **Ingest** and **sync** use the 202 Accepted pattern: `POST` returns a `task_id` immediately; `GET /ingest/{task_id}` or `GET /sync/{task_id}` returns `running`/`completed`/`failed` status.
- **Sync** rejects concurrent requests with HTTP 409 because a second sync against the same registrations would race on LanceDB writes.
- **Ingest** allows concurrent requests — each operates on independent documents and gets its own task.
- **Remember** remains synchronous on HTTP because inline content is already in memory; no network fetch is required. If embedding latency becomes a concern for very large inputs, it should follow the same 202 pattern.
- **Optimize** is CLI-only (no HTTP or MCP surface). It runs LanceDB compaction and index rebuilds, which are CPU-intensive and should not be triggered remotely without explicit intent.

A.3 Task Lifecycle

Background tasks (ingest, sync) follow the same lifecycle:

1. `POST` validates input, creates a `TaskState` record on the server context, spawns an `asyncio.Task` wrapping `run_in_threadpool()`, and returns HTTP 202 with `task_id` and `status` = `accepted`.
2. The background task updates `status` to `completed` (with `results`) or `failed` (with `error`) on completion.
3. A `finally` guard marks any task that exits without setting a terminal status as `failed`.
4. `CancelledError` is caught before `Exception` to ensure clean shutdown on server stop.
5. The CLI prints the `task_id` and exits 0 (fire-and-forget).